Student Expense Predictor using Machine Learning

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1. INTRODUCTION

In today's modern student life, managing expenses has become a major challenge. Students often spend money on daily needs such as food, travel, and entertainment, but they do not track or predict their monthly expenses effectively. As a result, they face financial imbalance and difficulty in budgeting.

Artificial Intelligence (AI) and Machine Learning (ML) provide powerful techniques to analyse data and make predictions.

In this project, we apply Machine Learning to predict a student's monthly expenses based on their daily spending habits.

This project demonstrates how simple ML models can solve real-world problems efficiently.

2. PROBLEM STATEMENT

Students do not have a proper system to estimate their monthly expenses based on their daily spending patterns.

This lack of prediction leads to overspending and poor financial planning.

3. OBJECTIVE

The main objectives of this project are:

- To build a system that predicts monthly expenses
- To apply Machine Learning concepts from the syllabus
- To understand how data affects predictions
- To demonstrate real-world application of AI/ML

4. REAL-WORLD APPLICATION

This project can be useful in:

- Student budgeting systems
- Financial planning apps
- Expense tracking applications
- Personal finance management tools

5. MACHINE LEARNING CONCEPT USED

Linear Regression

Linear Regression is a supervised learning algorithm used to predict continuous values.

Why Linear Regression?

- Simple and easy to implement
- Works well for numerical prediction
- Directly part of syllabus (CO4)

Concept:

It finds a relationship between:

```
Input variables → Output variable
```

Example:

```
More daily spending → Higher monthly expense
```

6. DATASET DESCRIPTION (data.csv)

The dataset is manually created and contains the following columns:

Data Set	Description
daily_expense	Average daily spending
outings	Number of outings per month
food_cost	Monthly food expenses
travel_cost	Monthly travel expenses
total_expense	Total monthly expense (target)

Example Data:

</> csv

daily_expense,outings,food_cost,travel_cost,total_expense

200,5,3000,1000,9000

```csv
daily_expense,outings,food_cost,travel_cost,total_expense
200,5,3000,1000,9000
```

7. SYSTEM ARCHITECTURE

```
Data (CSV)
   ↓
Data Processing (Pandas)
   ↓
Model Training (Linear Regression)
   ↓
Model Saving (Pickle)
   ↓
User Input
   ↓
Prediction
   ↓
Output (Expense)
```

8. FILE STRUCTURE

```
expense-predictor/
│
├── data.csv
├── train_model.py
├── predict.py
├── model.pkl
├── README.md
```

9. LIBRARIES USED

1. pandas

Purpose:

- Used for handling and analyzing data

Functions Used:

- read_csv() → Reads CSV file

2. scikit-learn (sklearn)

Purpose:

- Machine Learning library

Module Used:

- LinearRegression

Functions:

- fit() → trains model
- predict() → makes prediction

3. pickle

Purpose:

- Used to save and load model

Functions:

- dump() → save model
- load() → load model

10. CODE EXPLANATION

Load dataset

Python

```python
data = pd.read_csv("data.csv")
```

Reads CSV file into table format

Define inputs

Python

```python
X = data[['daily_expense', 'outings', 'food_cost', 'travel_cost']]
```

Features (independent variables)

Define output

Python

```python
y = data['total_expense']
```

Target (dependent variable)

Train model

Python

```python
model = LinearRegression()
model.fit(X, y)
```

Model learns patterns from data

Save model

Python

```python
pickle.dump(model, open("model.pkl", "wb"))
```

Saves trained model

predict.py

Load model

Python

```python
model = pickle.load(open("model.pkl", "rb"))
```

👉 Loads trained model

Take input

Python

```python
daily = float(input("Daily expense: "))
```

👉 Takes user input

Create structured data

Python

```python
input_data = pd.DataFrame(...)
```

👉 Converts input into correct format

Predict

Python

```python
model.predict(input_data)
```

👉 Generates output

🔹 **Display result**

Python Run

```python
print(result)
```

👉 Shows prediction

11. WORKING PROCESS

1. Load dataset
2. Train model
3. Save model
4. Take user input
5. Convert input
6. Predict output
7. Display result

12. RESULTS

The system successfully predicts monthly expenses based on input values. The predictions are close to expected values based on training data.

13. LIMITATIONS

- Dataset is small
- Prediction accuracy depends on data quality
- Does not consider unexpected expenses

14. FUTURE SCOPE

- Use real-world dataset
- Add graphical interface
- Add mobile app support
- Improve accuracy using advanced models

15. CONCLUSION

This project demonstrates the application of Machine Learning to solve a real-world problem. Using Linear Regression, we successfully built a system that predicts student expenses. The project highlights how simple AI techniques can be used effectively in daily life.